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Luca Gianni has worked during his career on new drug development since phase I trials and on the definition of innovative drug therapies in phase II and phase III trials. Since 1995, he has focused on clinical and translational research in breast cancer, contributing to the definition and understanding of HER2-directed therapies in the neoadjuvant setting.

Rakesh K. Jain is the Andrew Werk Cook Professor of Radiation Oncology (Tumour Biology) at Harvard Medical School and Director of the Edwin L. Steele Laboratories at Massachusetts General Hospital (USA). He is widely known for uncovering how the abnormal tumour microenvironment fuels tumour progression and confers resistance to molecular nanomedicines and immunotherapies, developing innovative strategies to 'normalize' the tumour microenvironment, and then translating these strategies from bench to bedside. An author of 800 publications, he is among the top 1% cited researchers in clinical medicine.

Guido Kroemer is Director of the Metabolism, Cancer & Immunity laboratory in Paris (France). His team discovered the contribution of immunosurveillance to the success of cancer chemotherapy, radiotherapy and targeted therapy. He also launched an 'ecological' theory of malignant disease that underscores the decisive impact of bodywide alterations on the progression and therapeutic response of cancer. He is currently President of the European Academy of Tumour Immunology as well as of the European Network of Cancer Immunotherapy.

Josep M. Llovet is Professor of Medicine and Director of the Liver Cancer Program at Mount Sinai (USA), Professor of Medicine-Hepatic Oncology at the University of Barcelona – Hospital Clinic (Spain), and ICREA Research Prof at IDIBAPS (Spain). He has published ~360 papers and is the world's most-cited investigator in liver cancer (>140,000 citations).

Georgina V. Long is co-medical director of Melanoma Institute Australia, and Professor of Melanoma Medical Oncology and Translational Research at the University of Sydney and Royal North Shore Hospital (Sydney, Australia). She leads an extensive clinical trials team and translational laboratory, with a focus on immuno-oncology in melanoma, including the neoadjuvant platform and treatment of brain metastasis.

Klaus Pantel is the founder (2002) and chairman of the Institute of Tumour Biology at the University Medical Center Hamburg-Eppendorf (Germany). He graduated in 1986 and finished his MD thesis on mathematical modelling of haematopoiesis. After his postdoctoral period in the USA (Wayne State University, Detroit, USA) in experimental haematology and stem cell research, he performed research in tumour immunology at the Institute of Immunology, University of Munich (Germany) for 10 years.

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Howard I. Scher was Chief of the Genitourinary Oncology Service at MSKCC for 25 years and currently serves as Head of MSKCC's Biomarker Development Program. He has dedicated his career to the co-development of targeted therapies and biomarkers and has led international efforts to standardize the design and analysis of prostate cancer clinical trials. He has received awards, including the AACR Team Science Award, the Donald S. Coffey-Prostate Cancer Foundation Physician Scientist award and the 2023 Giants of Cancer Care in Genitourinary Cancer.

Two decades of advances in clinical oncology lessons Takeda, Department of Medical Oncology, Vall d'Hebron Hospital Campus and Director of the Vall d'Hebron Institute of Oncology (Spain) and the current Chair of Cancer Core Europe. He has served as Principal Investigator of several phase I pharmacodynamic studies and translational projects with tumour-directed targeted therapies and cancer immunotherapy.

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Yi-Long Wu is a tenured Professor of Guangdong Provincial People's Hospital, Guangdong Lung Cancer Institute (China). He is the President of Chinese Thoracic Oncology Group, the Past President of Chinese Society of Clinical Oncology (2013–2017) and member of the Board of Directors of the International Association Study of Lung Cancer from 2013 to 2017. He received the International Association Study of Lung Cancer Science award in 2015 and Highly Cited Researcher award in the field of Clinical Medicine 2018–2023.

Competing interests

S.B. has received institutional funding for academic studies from AstraZeneca and GSK; honoraria for lectures from Abbvie, AstraZeneca, GSK, ImmunoGen, Merck Sharpe & Dohme, Mersana, Novacure, Pfizer, Roche, Takeda and Verastem Oncology; has had advisory roles for Abbvie, AstraZeneca, Epsilon, GSK, Immunogen, ITM Oncologics, Merck Sharpe & Dohme, Mersana, Myriad, Novartis, Onxerna, Regeneron, Seagen, Shattuck Labs, TORL Biotherapeutics, Verastem Oncology and Zymeworks; received travel support from AstraZeneca, GSK and Verastem Oncology; and is a past Director of Membership at ESMO (2020–2022). A.D. has received honoraria from or had advisory roles for 14ner/Elevation Oncology, Amgen, Anheart Therapeutics, Abbvie, ArcherDX, AstraZeneca, Bayer, Beigene, BergenBio, Blueprint Medicines, Boundless Bio, Bristol Myers Squibb, Chugai Pharmaceutical, EcoR1, EMD Serono, Entos, Exelixis, Helsinn, Hengrui Therapeutics, Ignyta/Genentech/Roche, Innocare, Janssen, Loxo/Lilly, Merus, Monopteros, MonteRosa, Novartis, Nuvalent, Pfizer, Prelude, Regeneron, Repare RX, Takeda/Ariad/Millennium, Treeline Bio, TP Therapeutics, Tyra Biosciences, Verastem Oncology and Zymeworks; has received institutional funding from Foundation Medicine, GSK, Pharmamar, Taiho and Teva; owns equity in mBrace and Treeline; has a copyright for selpercatinib-osimertinib (pending, US 18/041,617); has received royalties from UpToDate and Wolters Kluwer; has received support for food/beverage from Boehringer Ingelheim, Merck and Puma; and has received continuing medical education-related honoraria from Answers in CME, Applied Pharmaceutical Science, AXIS, Clinical Care Options, Doc Congress, EPG Health, Harborside Nexus, I3 Health, Imedex, Liberum, Med Learning, Medendi, Medscape, MedTalks, MJH Life Sciences, MORE Health, Ology, OncLive, Paradigm, Peervoice Institute, PeerVoice, Physicians Education Resources, Projects in Knowledge, Resources, Remedica, Research to Practice, RV More, Springer Healthcare, Targeted Oncology, TouchIME and WebMD. T.J.F. has patents in the field of immunotherapy. I.M.G. has had advisory roles for GSK, Janssen, Menarini, Pfizer and Sanofi; and is a co-founder of Predicta bio. L.G. has had advisory roles for Artemida Pharma, AstraZeneca, Beigene, Daiichi-Sankyo, Pfizer, Roche and Zymeworks; has had consulting roles for selected programmes of Amgen, Biomedical Insights, Denali Therapeutics, Menarini, Revolution Medicines, Synaffix and Ymeworks; and is a co-inventor of European Patent Application N. 12195182.6 and 12196177.5 titled “PDL-1 expression in anti-HER2 therapy” with Roche (no compensation provided). R.K.J. has received consultant fees from DynamiCure and SynDevRx; owns equity in Accurius, Enlight and SynDevRx; and has served on the boards of trustees of Tekla Healthcare Investors, Tekla Healthcare Opportunities Fund, Tekla Life Sciences Investors and Tekla World Healthcare Fund; and received grants from Boehringer Ingelheim and Sanofi. G.K. has held research contracts with Daiichi-Sankyo, Eleor, Kaleido, Lytix Pharma, PharmaMar, Osasuna Therapeutics, Samsara Therapeutics, Sanofi, Sutro, Tollis and Vasage; is on the board of directors of the Bristol Myers Squibb Foundation France; is a scientific co-founder of everImmune, Osasuna Therapeutics, Samsara Therapeutics and Therafast Bio; is in the scientific advisory boards of Hevolution, Institut Servier, Longevity Vision Funds and Rejuvenon Life Sciences; and has patents covering therapeutic targeting of aging, cancer, cystic fibrosis and metabolic disorders, of which “Methods for weight reduction” (US11905330B1) is relevant to this study. G.K.'s spouse has held research contracts with 9m, Daiichi-Sankyo, GSK, Incyte, Innovate Pharma, Lytix, Kaleido, Merus, Pilege, Roche, Transgene, and Tusk and Roche; was on the board of directors of Transgene; is a co-founder of everImmune; and holds patents covering the treatment of cancer and the therapeutic manipulation of the microbiota. J.M.L. has received research support from Bayer HealthCare Pharmaceuticals and Eisai; and consulting fees from AstraZeneca, Bayer HealthCare Pharmaceuticals, Bristol Myers Squibb, Eisai, Genentech, Glycotest, Merck, Moderna and Roche. G.V.L. is a consultant adviser for Agenus, Amgen, Array Biopharma, AstraZeneca, Bayer, BioNTech, Boehringer Ingelheim, Bristol Myers Squibb, Evaxion, Hexal AG (Sandoz Company), Highlight Therapeutics, IOBiotech, Immunocore, Innovent Biologics USA, Merck Sharpe & Dohme, Novartis, PHMR, Pierre Fabre, Regeneron, Scancell and SkylineDX. 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J.T. has had scientific consultancy roles for Alentis Therapeutics, AstraZeneca, Aveo Oncology, Boehringer Ingelheim, Cardiff Oncology, CARSGen Therapeutics, Chugai, Daiichi-Sankyo, F. Hoffmann-La Roche, Genentech, hC Bioscience, Immodulon Therapeutics, Inspima, Lilly, Menarini, Merck Serono, Merck Sharpe & Dohme, Merus, Mirati, Neophore, Novartis, Ona Therapeutics, Ono Pharma USA, Orion Biotechnology,

learned and future directions

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Peptomyc, Pfizer, Pierre Fabre, Samsung Bioepis, Sanofi, Scandion Oncology, Scorpion Therapeutics, Seattle Genetics, Servier, Sotio Biotech, Taiho, Takeda Oncology and Tolremo Therapeutics; holds stock at 1TRIALSP, Alentis Therapeutics, Oniria Therapeutics and Pangaea Oncology; and has had educational collaborations with Medscape Education, PeerView Institute for Medical Education and Physicians Education Resource. R.R.W. is a co-founder and Chief Scientific Officer at PersonaDX (USA), which includes start-up stock but no other income. M.W. has received research grants from Novartis, Quercis and Versameb; honoraria for lectures or advisory board participation or consulting from Anheart, Bayer, Curevac, Medac, Neurosense, Novartis, Novocure, Orbus, Pfizer, Philogen, Roche and Servier; and is a board member at the European Organisation for Research and Treatment of Cancer. Y.L.W. has received institutional research support from AstraZeneca, Boehringer Ingelheim, Bristol Myers Squibb, Hengrui and Roche; speaker fees from AstraZeneca, Beigene, Boehringer Ingelheim, Bristol Myers Squibb, Eli Lilly and Company, Merck Sharp & Dohme, Pfizer, Roche and Sanofi; and has unpaid memberships in governmental organizations of ASCO, ESMO, International Association Study of Lung Cancer, Chinese Society of Clinical Oncology and Chinese Thoracic Oncology Group. C.M.B., E.B., M.W.B. and K.P. declare no competing interests.

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Abstract

Since the publication of the first issue of *Nature Reviews Clinical Oncology*, we have witnessed advances in multiple research areas that have culminated in improved outcomes for many cancer types, although substantial unmet needs remain for a majority of patients worldwide. Here, we have asked experts in several key specialities to reflect on the progress from the past 20 years and propose the next steps to enable further advances. Although we are aware that this Viewpoint cannot provide full coverage of the vast field that is clinical oncology, we hope that these messages inspire a diverse range of readers.

Over the past 20 years, what have been the main advances in your area of research interest?

Guido Kroemer:

Twenty years ago, cancer was viewed as a disease of (epi)genomically unstable, ever-more selfish malignant cells¹. However, over the past two decades, all major stakeholders in oncology, including clinicians, researchers and drug developers, have understood that cancer only develops when immunosurveillance fails. Thus, to thrive, neoplastic cells must 'trick out' immune cells or develop in a context in which the immune system is already affected by hereditary or acquired defects. This notion has deeply affected biomarker discovery, which is now dominated by the search for determinants of cancer immunogenicity

and deep immunophenotyping of tumour-infiltrating leukocytes. In addition, mechanistic insights on immunosurveillance have led to the development of an expanding arsenal of immunotherapies that is steadily conquering the realm of clinical oncology.

Rakesh K. Jain:

Despite the importance of the tumour microenvironment (TME) — the ‘soil’, as defined by Steven Paget in 1889 — in cancer progression, oncology research has been largely focused on cancer cells — the ‘seed’. The past 20 years have witnessed major advances in our understanding of the TME and the mechanisms by which it mediates tumour progression and resistance to treatment. A major example of such an advance is the reprogramming of the immune TME using immune-checkpoint inhibitors (ICIs), a discovery that has revolutionized the treatment of multiple solid tumour types and was recognized with the 2018 Nobel Prize in Physiology or Medicine to James Allison and Tasuko Honjo. Another advance is the recognition and translation of the counterintuitive concept that vascular normalization using anti-angiogenic agents (which were originally developed to kill tumour vessels) to mitigate deleterious aspects of the TME can improve the response of solid tumours to radiation, cytotoxic agents and/or ICIs². As a result, vascular-normalizing drugs targeting the VEGF–VEGFR pathway, alone and combined, are now part of FDA-approved treatment regimens for many solid tumour types³. Finally, mathematical modelling and machine learning approaches are providing deeper mechanistic insights, aiding drug discovery and design, and being tested for cancer detection and diagnosis.

Georgina V. Long:

For as long as immunologists have been unveiling the complexity, capability and reach of the human immune system, oncologists have been trying to direct this power against cancer. In 2008–2012, we finally had breakthrough success with the development of clinical ICIs. Prior to the introduction of these agents, (chemorefractory) cancers with distant metastases were almost universally fatal. Now, ICIs are providing durable cancer control to over half of patients with stage IV melanoma⁴ and are drastically reducing the risk of post-surgical recurrence in those with stage III melanoma⁵ — these are the most important advances that the field has ever seen.

A crucial aspect of the treatment of patients with melanoma using ICIs has been learning how to elicit the best outcomes for tumour elimination while avoiding toxicity by integrating insights on tumour–immune biology into trial design. One example is the randomized phase III NADINA trial involving patients with stage III melanoma, in which a simple shift in the timing of drug delivery to precede (neoadjuvant) rather than follow (adjuvant) surgery⁶ reduced the risk of progression presurgery, recurrence post-surgery or death from melanoma by 68%⁵. Compared with the situation from 5 years ago, when patients did not receive perioperative therapy, this improvement is equivalent to a >80% reduction in recurrence risk. Emerging from investigator-led initiatives, trials testing neoadjuvant ICIs reflect the power of data-driven, clinician-led research. Other key insights include the need to optimize ICI combinations and shift away from chemotherapy-informed models of drug delivery towards flexible therapy intervals that avoid toxicity by synchronizing treatment administration with immune activation in each individual.

Yi-Long Wu:

ICIs have revolutionized the management of patients with non-small-cell lung cancer (NSCLC) and small-cell lung cancer (SCLC) without targetable alterations. These therapeutic agents have significantly improved survival outcomes for patients with lung cancer. In those with advanced-stage NSCLC, anti-PD-1 or anti-PD-L1 antibodies have increased 5-year overall survival (OS) from <10% to ~20%. Patients with unresectable stage III NSCLC receiving the anti-PD-L1 antibody durvalumab as consolidation therapy after chemoradiotherapy have a remarkable 5-year OS of 42.9%. In those with resectable NSCLC, neoadjuvant, adjuvant and perioperative administration of ICIs has led to impressive OS outcomes, reducing the risk of death by 28% in adjuvant trials. In patients with SCLC, the addition of ICIs to chemotherapy prolongs median OS (15.4 months versus 10.9 months with chemotherapy alone)⁷.

Research conducted in China has had a major contribution towards ICIs becoming a standard-of-care (SOC) treatment for patients with lung cancer, particularly those with SCLC. These studies have provided new agents and trial evidence, shaping the landscape of immunotherapy for lung cancer. With localized NSCLC becoming increasingly curable and advanced-stage disease transitioning to a chronic state, the past two decades have witnessed substantial progress⁸.

Terry J. Fry:

After two decades of research during which initial clinical experience was met with limited success, in the 2010s, the adoptive transfer of T cells genetically modified to express chimeric antigen receptors (CARs) demonstrated remarkable efficacy in a number of parallel clinical trials involving adults and children with haematological malignancies^{9,10}. Along with several other modifications to the initial prototype CARs, the major design change that contributed to this improved efficacy was the incorporation of a costimulatory signalling domain in line with the primary T cell activation domain. Based on complete remission induction rates of 80% in children with highly refractory acute lymphoblastic leukaemia and on complete response rates of 50% in adults with lymphoma refractory to three or more lines of prior therapy, CAR T cells targeting the B cell antigen CD19 moved quickly from academic trials into industry-led registration trials. These cellular products were approved by the FDA in 2017, first for paediatric acute lymphoblastic leukaemia, based on the results from remarkably small trials owing to the extremely high efficacy in patients with refractory disease^{9,10}. In addition to the fact that approvals for paediatric patients preceded those for adults, other notable aspects of these regulatory decisions include the similar objective response rates across multiple trials testing different versions of the CD19 CAR construct, the fact that this was the first gene therapy approval and the ability for the pharmaceutical industry in collaboration with regulators to develop manufacturing infrastructure for the routine generation of patient-specific treatments. These approvals also set the clinical and regulatory foundation for the field of cell and gene therapy in cancer more broadly. The success of CD19-targeted CAR T cells was followed by a rapid proliferation of preclinical and clinical studies that resulted in the approval of B cell maturation antigen-targeting CAR T cells for patients with multiple myeloma.

Irene M. Ghobrial:

In the past two decades, treatment options for patients with multiple myeloma and the resulting survival outcomes have improved substantially. The approval of >20 new drugs, including novel therapies and immunotherapies, has increased the median OS from 3–5 years to 10–15 years¹¹. Progress in biomarker research, genomics, and understanding of clonal evolution and the role of the immune system has been pivotal to these developments¹². Immunotherapies, such as bispecific T cell engagers and CAR T cells, have emerged in the past decade, leveraging the immune system against myeloma cells¹³. Identifying minimal residual disease (MRD) as a potential surrogate end point for survival has hastened the development of new therapies.

Despite these advances, multiple myeloma remains incurable, with many patients dying from the disease or treatment-related complications. Diagnosis often occurs after symptoms appear, despite the availability of blood biomarkers for early detection (such as measurement of monoclonal proteins in blood). Patients with high-risk cytogenetic features often have disease resistant to treatment, including immunotherapies. Individuals from ethnic minorities, particularly Black individuals, face disparities in myeloma care, including delayed diagnosis and limited access to novel therapies and clinical trials, leading to inferior survival relative to white individuals. Access to costly treatments is also unequal globally. Finally, prolonged treatment with multidrug regimens can lead to severe toxicities, reducing quality of life (QOL).

Alexander Drilon:

This year's celebration of the twentieth anniversary of the discovery of the role of *EGFR* mutations in lung cancer¹⁴ reminds us that the past two decades have encompassed a 'Renaissance period' for targeted therapies in oncogene-driven lung cancers. The reality that multiple targeted therapies for 11 distinct classes of oncogenic alterations have been approved is astounding progress. Iterative drug design improved on early-generation tyrosine-kinase inhibitors (TKIs), resulting in more-selective and central nervous system-penetrant next-generation TKIs that abrogated kinase-intrinsic resistance. Such next-generation agents not only provide opportunities for sequential targeted therapy but also replaced early-generation TKIs as frontline treatments for patients with metastatic NSCLC. Novel approved therapies have also diversified mechanistically and moved beyond TKI monotherapy to include GTPase inhibitors, bispecific antibodies, antibody–drug conjugates (ADCs), and combinations of targeted agents and chemotherapy. Finally, adjuvant therapy with targeted agents has been approved for patients with non-metastatic oncogene-driven NSCLCs, marking a much-needed move into the treatment of earlier-stage disease.

Luca Gianni:

With the current abundance of drugs for patients with breast cancer, selection of the least intensive and most efficacious therapy for each individual patient is crucial — neoadjuvant treatment approaches can address this need. Pathological complete response and residual disease after neoadjuvant treatment are valid surrogate end points of individual clinical benefit and provide formidable guidance for subsequent treatment decisions. Landmark examples include: (1) the KATHERINE trial¹⁵, which showed better long-term efficacy (OS

and disease-free survival (DFS)) in patients with HER2-positive breast cancer and residual disease after neoadjuvant trastuzumab with or without pertuzumab when treatment was switched to the ADC trastuzumab emtansine; (2) the CREATE-X UMIN trial¹⁶, in which detection of residual disease after neoadjuvant chemotherapy prompted the use of adjuvant capecitabine, resulting in significantly improved DFS in patients with triple-negative breast cancer; and (3) the POETIC trial¹⁷, which showed that Ki67 assessment at surgery after 2 weeks of neoadjuvant endocrine therapy can enable prediction of clinical benefit from endocrine therapies in patients with oestrogen receptor-positive breast cancer.

Josep Tabernero:

In gastrointestinal oncology, we have celebrated a myriad of advances over the past two decades, including the development of new treatment strategies, molecular markers and diagnostic tools, redefining the multidisciplinary management and treatment of patients across many gastrointestinal cancer types. Genetic and transcriptomic analyses have spurred more effective therapies matched to distinct molecular subtypes.

The development of targeted drugs for patients with advanced-stage and metastatic disease has resulted in a paradigm shift from prolonging survival to cure. In the past decade, ICIs have changed the treatment landscape within different patient subgroups, including those with microsatellite instability.

Adding to the burden of gastrointestinal cancers, which account for a quarter of all cancers and a third of cancer-related deaths globally¹⁸, the incidence of several cancer types is on the rise, particularly that of colorectal cancer (CRC). Sadly, by 2030, CRC could become the deadliest cancer in young adults (20–49 years of age)¹⁹. We need to tackle the unthinkable scale and associated implications of this challenge now.

Howard I. Scher:

Over the past two decades, 16 new drugs have been approved by the FDA, revolutionizing the therapeutic landscape for prostate cancer and improving patient survival. However, understanding of the biology of each patient's disease has been limited; as a result, treatment decisions have largely focused on the androgen receptor or androgen receptor signalling axis followed by cytotoxic drugs. In the past decade, the increasing capability of tumour profiling has increased biology-guided treatment selection²⁰. Examples include poly(ADP-ribose) polymerase (PARP) inhibitors and ICIs, both of which now have formal FDA-approved indications. Current practice also includes PET-based imaging using tracers targeting prostate-specific membrane antigen (PSMA) to detect disease and identify patients likely to benefit from ¹⁷⁷Lu-PSMA-directed theranostic agents, which are also approved by the FDA and widely used along with fluorodeoxyglucose-based imaging to identify areas of active versus inactive disease.

Taken together, the broad availability of therapeutic options has improved the outcomes of patients with prostate cancer. However, all disease is only eliminated in a few of them, which can lead to the emergence of new disease entities owing to lineage plasticity, a transition from adenocarcinoma to a neuroendocrine phenotype consisting of markedly heterogeneous cell populations^{21,22}. Most of these populations are not androgen receptor driven, lack

androgen receptor or PSMA expression, and reside in individual sites of metastatic dissemination that are biologically diverse to the point of needing to be considered and approached therapeutically as ‘individual patients’. Patients with this neuroendocrine subtype have typically received platinum-based therapies and have no SOC treatment options available because their disease cannot be considered as a single entity.

Josep M. Llovet:

Systemic therapies have completely revolutionized the management of patients with hepatocellular carcinoma (HCC). Around 60% of these patients are diagnosed or will progress to advanced-stage disease with a short life expectancy (median natural history OS of ~8 months). The advent of the multikinase inhibitor sorafenib in 2008 (ref. 23) represented the dawn of a new era of molecularly targeted therapies providing clinical benefit, epitomized by small but statistically significant improvements in OS in patients receiving TKIs both in the first line (sorafenib and lenvatinib; median OS of 13–19 months) and second line of treatment (regorafenib, cabozantinib and ramucirumab; median OS of 9–11 months). In the past few years, a second wave of advances emerged when ICI-containing regimens were shown to further improve OS in the first line of treatment (median OS of ~16–23 months), QOL and objective response rates (up to 30–36%). The combination of atezolizumab plus the anti-VEGF antibody bevacizumab²⁴ was the first regimen improving the outcomes of patients relative to sorafenib, whereas subsequent dual ICI combinations (such as durvalumab plus tremelimumab or nivolumab plus ipilimumab) have led to long-lasting responses, with ~25% of patients alive at 3–4 years.

Susana Banerjee:

Maintenance therapy with PARP inhibitors has revolutionized the management of patients with ovarian cancer, initially in those with mutations in *BRCA1* and/or *BRCA2* and then in patients with homologous recombination deficiency and without such alterations, in both the recurrent and newly diagnosed setting. The sustained progression-free survival (PFS) improvement in patients with newly diagnosed, *BRCA1/2*-mutated advanced-stage ovarian cancer (56 months versus 13 months with placebo)²⁵ provided hope for the possibility of long-term remission.

In the past few years, ICIs have transformed clinical practice and, for the first time in decades, patients with recurrent and/or metastatic endometrial cancer²⁶ and cervical cancer both have treatment options with superior efficacy over chemotherapy alone in the first-line setting. Undoubtedly, human papillomavirus vaccination and screening have been the most effective strategies for reducing the burden of cervical cancer worldwide. For patients with recurrent and/or metastatic disease, the addition of bevacizumab and then pembrolizumab to first-line chemotherapy and second-line tisotumab vedotin as monotherapy, the first approved ADC for cervical cancer, have led to a stepwise improvement of OS. Mirvetuximab soravtansine is the first ADC approved for patients with ovarian cancer and the first biomarker-directed therapy (for folate receptor α -high disease) shown to improve OS in platinum-resistant ovarian cancer²⁷.

Markus W. Büchler:

Pancreatic cancer has the worst prognosis of all human cancers and, unfortunately, the latest advances in precision medicine and immunotherapy have not improved survival rates for this disease. Historically, surgery has been the only curative approach for patients with pancreatic cancer; however, throughout the twentieth century, the overall outcomes after resection were poor, with a 5-year OS of 10% at best. This situation changed in 2004 with results from the pivotal ESPAC-1 trial demonstrating that, in the hands of specialized surgeons and when surgery is combined with adjuvant chemotherapy (but not chemoradiotherapy), patients can have a long-term (5-year) OS of 20%²⁸. Adjuvant combination chemotherapies, such as gemcitabine–capecitabine and modified 5-fluorouracil, irinotecan and oxaliplatin (mFOLFIRINOX) tested in the ESPAC-4 and PRODIGE-24 trials, respectively, have further improved 5-year OS to 30–50%. In the past few years, neoadjuvant combination chemotherapy has been shown to increase resectability rates and OS in patients with borderline resectable or locally advanced pancreatic cancer²⁹.

Michael Weller:

The past 20 years have seen major developments in the characterization of the molecular genetic evolution of primary brain tumours that have refined their classification as reflected by several revisions of the WHO classification (the last published in 2021 (ref. 30)). Molecular biomarkers are now essential to assigning lesions to distinct tumour entities, and profiling the DNA methylome has become a powerful tool that aids in brain tumour classification. In parallel, the tools used for neuroimaging of brain tumours have evolved with the addition of multiple new MRI and PET techniques, with focus on predicting the biological behaviour of tumours — notably their response to therapy. Artificial intelligence is beginning to be introduced for the imaging assessment of primary brain tumours.

Aided by improved imaging, surgical approaches have evolved based on several techniques, including neuronavigation, ultrasonography and fluorescent labelling. The extent to which these advances in surgical techniques have improved OS remains controversial, although surgery has undoubtedly become much safer and is associated with much less morbidity and mortality than it was decades ago.

In parallel with the improved classification of brain tumours, targeted pharmaceutical interventions have become an option or even the SOC, albeit only for a minority of tumours. This advance is particularly relevant in the paediatric patient population³¹.

Kathy Pritchard-Jones:

In paediatric oncology, there have been advances in three main areas. First, we have seen a massive expansion in understanding of the underlying genetic make-up of paediatric cancers, with the discovery of hundreds of genetic alterations³². These distinctive molecular profiles are often not shared with adult cancers or can involve the same genes but affect different pathways or dependencies. This knowledge has enabled further definition of risk groups for treatment stratification and identification of a few predictive biomarkers for a limited number of targeted therapies. Compared with cancers in adults, paediatric cancers generally have a lower tumour mutational burden but share the phenomenon of

somatic heterogeneity with its implications for the interpretation of findings on molecular biomarkers and clonal evolution. Moreover, whole-exome and genomic sequencing have revealed that a higher proportion of children than previously thought have a genetic predisposition to cancer: 10% overall and a much higher percentage in certain subtypes.

Second, we have recognized the need for paediatric cancer-specific drug development approaches. The limitations of current efforts to incentivize industry to make novel targeted drugs developed for adult patients available to paediatric patients with the same genetic alterations have emphasized the need for a mechanism-of-action-based approach to drug development. Many targets are unique to paediatric cancers and involve signalling pathways controlling developmental processes, which have not yet been translated into targeted therapies. The creation of the ACCELERATE Platform has started to address these challenges³³. Of note, in some instances, such as the development of CAR T cell therapies, paediatric advances in academia have transformed therapeutic options for some refractory leukaemias ahead of their use in adults.

Finally, since the early 2000s, we have seen the consolidation of multinational clinical trial groups that run studies across entire continents, especially in Europe. The work of these groups has defined tumour-specific SOC treatments that are widely used.

Ralph R. Weichselbaum:

Radiation oncologists have been concerned mainly with the treatment of local and regional cancers. Combinations of radiation with cytotoxic agents and other systemic therapies (such as ICIs), and technical improvements in radiotherapy delivery and imaging have raised the therapeutic ratio in several tumour types by increasing local control and decreasing toxicity to non-malignant tissue.

The use of radiotherapy in patients with metastatic cancers, which account for most cancer-related mortality, has traditionally focused on symptom palliation. An important area currently under investigation is the integration of radiotherapy into curative treatments for these patients. Most of the interest has focused on the oligometastatic hypothesis, which states that cancers are not always widely metastatic but exist as a spectrum of disease both in the pace of progression and in the number and sites of metastasis³⁴. A corollary to this hypothesis is that radiotherapy — and other ablative therapies — might be curative in some patients and contribute to cure in others: robust responses to systemic treatments might be improved by ablating remaining resistant tumour clones (or ‘oligoprogressors’)³⁵. Randomized clinical trials (RCTs) have tested radiotherapy with or without systemic therapies in patients with oligometastatic and oligoprogressive disease, with the majority demonstrating improvements in OS; however, two phase III trials have been negative for OS.

Klaus Pantel:

The term liquid biopsy was introduced in 2010 to denote the analysis of circulating tumour cells in the blood of patients with cancer³⁶ and was rapidly adopted to refer to tumour-derived products soluble in body fluids, especially cell-free circulating DNA (ctDNA) fragments. Liquid biopsy is a new diagnostic concept developed in the past two decades that enables obtaining tumour-specific information in patients from whom tumour tissue

cannot be obtained or is difficult to obtain safely. Needle biopsies are invasive procedures and some locations are not easy to access, which limits the use of this gold standard in cancer diagnostics. By contrast, drawing blood is a minimally invasive procedure and blood is a pool of tumour cells or other analytes released from all possible locations (albeit with different shedding rates). Over the past 10 years, ctDNA has received enormous attention as a liquid biopsy-based biomarker, while research on circulating tumour cells has opened new avenues for a better understanding of tumour biology in patients with cancer, including inpatient heterogeneity³⁷.

Eduardo Bruera:

Patients with advanced-stage cancer develop devastating physical, psychosocial, spiritual and financial distress. The dimensions and intensity of suffering can vary substantially between patients with the same type and stage of cancer. Palliative care emerged as a speciality in the 1960s and 1970s, preceding the establishment of medical oncology. Although most of the care and research at that time was focused on patients with cancer, these programmes were developed outside of major cancer centres and universities. For decades, patients with cancer accessed palliative care only after disease progression on treatment, usually in the last days of life³⁸. During the 1990s, a few academic palliative care centres recognized the need to support patients much earlier in their disease trajectory. This approach involved moving care to outpatient centres and generating evidence from clinical trials to support successful integration of palliative care within routine management. Conducting such clinical trials was challenging owing to the limited number of academic palliative care programmes and meagre support from governments, philanthropy and industry.

The past 20 years have seen explosive growth in the evidence supporting integration of palliative care from the early signs of advanced-stage cancer. RCTs have demonstrated that specialist palliative care can improve physical and psychosocial outcomes, end-of-life care metrics and financial outcomes, with neutral or positive effects on OS^{38,39}. Universal access to palliative care for patients with advanced-stage cancer is now part of all major cancer guidelines^{39,40}.

Christopher M. Booth:

In the past two decades, some of the most important work in outcomes research has been related to defining the concept of ‘value’ in cancer care. Value represents the interface between benefit (the numerator) and cost (the denominator). By the mid-2000s, growing attention was placed on the rapidly rising price of oncology medicines (denominator). The lack of sustainability of these expenditures remains highly relevant in 2024 and presents major barriers to delivering high-quality and equitable care⁴¹.

In more recent years, interest in the numerator of the value construct has been growing. Broadly speaking, this focus relates to the validity of primary end points used and the magnitude of benefit observed in clinical trials. Since 2010, we have witnessed a major trend whereby the primary end point of oncology RCTs has shifted away from OS towards PFS and DFS⁴². These end points were initially considered as potential accurate predictors of

gains in OS (or surrogates); however, we have since learned that true surrogacy exists in only a limited number of clinical circumstances. In addition, the fact that measurement of these end points is vulnerable to informative censoring and other forms of bias is increasingly recognized. Although some novel anticancer medicines offer very large improvements in outcome (with major gains in OS), aggregate data reveal that median gains in PFS or OS in RCTs are modest, typically 2–3 months. By virtue of the efficacy–effectiveness gap, these gains are even smaller in routine practice. These limitations in contemporary medical oncology have tremendous relevance to patients who live with the clinical toxicity, time toxicity and financial toxicity of our treatments.

For the next 20 years, what should be the top priorities in your area of research?

G.K.:

Nowadays, clinical oncologists and their allies from the pharmaceutical industry tend to believe that the principal function of the immune system is anticancer immunosurveillance. Nevertheless, I argue that the most important mission of the immune system is the maintenance of tissue homeostasis, which is achieved by eliminating infectious pathogens (and often infected cells) as well as damaged or senescent cells in the context of ‘sterile inflammation’. Simultaneously, the immune system must avoid potentially harmful reactions against autoantigens, food-derived antigens and commensal microorganisms. This concept implies that several hitherto unsuspected external factors suppress anticancer immune responses. Thus, immunosurveillance is enfeebled by chronic autoinflammatory and autoimmune diseases, metabolic syndrome, ageing, and intestinal dysbiosis, all of which can occur independently from each other or — more often — in a compounded form. In this context, identifying actionable targets that might be suppressed to re-establish cancer immunosurveillance is of major importance. Such targets could include pro-inflammatory factors and cells, metabolites and (neuro)endocrine factors involved in metabolic syndrome, and signalling pathways that accelerate ageing and harmful microorganisms. The future of oncology involves a broad, holistic vision of the cancer–immune system crosstalk^{43,44}.

R.K.J.:

In the past 20 years, we have understood the importance of the TME and treatments targeting the non-cancer components of tumours, a research area that will continue to be exciting. A priority should be leveraging TME normalization and immunotherapies for pancreatic cancer, and primary or secondary brain tumours, for which effective therapies are still needed. The increased prevalence of obesity worldwide highlights the need for novel strategies to overcome the adverse consequences of this condition on the TME and response to treatment. Furthermore, leveraging insights into the role of physical forces (elevated interstitial fluid pressure and compressive forces)⁴⁵, microbiota, cancer cell–neuron crosstalk, diet and physical activity in cancer progression and response to treatment could lead to new therapeutic approaches. These priority areas will benefit from the judicious use of appropriate preclinical models of cancer, mathematical modelling and artificial intelligence.

Y-L.W.:

Over the next two decades, novel agents (such as bispecific ADCs) and new methods (such as personalized cancer vaccines) will emerge for patients with lung cancer. Refining patient selection, optimizing biomarker assessment and personalizing treatment will be pivotal to improving patient outcomes. Biomarker-guided adaptive treatment is gaining prominence; for example, several proof-of-concept trials have demonstrated the feasibility of ‘drug holidays’ guided by MRD assessment⁴⁶. After surgery or concurrent chemoradiotherapy, patients with lung cancer and undetectable MRD assessed through longitudinal monitoring could have prolonged OS without disease recurrence⁴⁷. These advances will rely on developments in liquid biopsy technologies. The integration of next-generation sequencing data with artificial intelligence models could increase the sensitivity, specificity and negative predictive value for detecting MRD to $\geq 90\%$, and patients cured from lung cancer could be identified more easily. Some phase III trials are exploring precision de-escalation or intensification of treatment strategies guided by MRD assessment. The goal is to maximize efficacy while minimizing unnecessary treatment and optimizing the use of medical resources.

G.V.L.:

Over the next 20 years, we must better understand the mechanisms of immunotherapy escape. In the melanoma field, the notion that patients with advanced-stage cancer can be cured with immunotherapies has become increasingly clear. By further optimizing the design, combination and delivery of immunotherapies — not just ICIs but also vaccines, bispecific T cell engagers, adoptive cell therapy (ACT) and more — to address resistance, we can unlock this curative potential in more patients.

T.J.F.:

As expected with a completely novel therapeutic modality, challenges with CAR T cell therapies have been identified. Relapse owing to therapeutic resistance (primarily loss or decrease of antigen expression levels) and CAR T cell failure (inadequate initial potency or short persistence) limit the durability of remissions to $< 50\%$ of treated patients⁴⁸. The success of CAR T cells in other types of cancer (in particular, solid tumours) has been very limited, owing to the lack of suitable antigens and TME-driven mechanisms of resistance, among other aspects. The final but perhaps most important challenge is ensuring access to this highly effective treatment for all patients who could benefit from it. Given the relative newness of this treatment approach and the remarkable progress that is being made across numerous fields (immunology, genome engineering or cell manufacturing techniques, to name a few) many of these challenges will probably be addressed over the next two decades, firmly embedding ACT into the treatment armamentarium for multiple cancer types.

I.M.G.:

For clinicians specialized in multiple myeloma, the primary goal should be cure rather than merely managing this malignancy as a chronic condition. Achieving this ambitious goal requires fundamental changes. First, we must deepen our understanding of disease biology, including the immune TME. To achieve cures, we must target and eradicate the

earliest progenitor cells of multiple myeloma. Second, early intervention is crucial. We need to detect and treat multiple myeloma at its earliest stages, long before symptoms appear. By then, the disease has often evolved substantially, leading to treatment resistance. Third, multiple myeloma should be treated as a spectrum of multiple diseases requiring tailored treatments rather than a singular entity. Such an approach involves assessing genetic and immune biomarkers early on to personalize therapy and avoid ineffective treatments.

Immunotherapies have shown promise in multiple myeloma and ongoing research is exploring their potential in individuals with precursor states or newly diagnosed disease. Advances in personalized vaccines, CAR T cell therapies and next-generation T cell engagers, including trispecific antibodies, could improve outcomes.

Finally, addressing ethnic disparities in cancer is essential and multiple myeloma constitutes a clear example of this challenge. Access to effective treatments, including immunotherapies, and clinical trials should be equitable. By pursuing these avenues, we can move closer to curing patients with multiple myeloma and enabling them to live cancer-free lives.

A.D.:

Although the number of ‘undruggable’ oncogenes continues to shrink, additional drug approvals — including treatments for rare alterations⁴⁹ — will rely on increased tumour-agnostic investigation, improved trial design (accounting for testing of combination therapies) and global stakeholder cooperation. Mechanistic diversification will remain crucial as certain oncogenic alterations might only be actionable with agents from therapeutic classes other than small-molecule kinase or GTPase inhibitors (such as protein degraders or epigenome modifiers, with therapeutic relevance that remains to be substantiated). Approaches targeting the proteome will continue to be a focus of research. This year’s tumour-agnostic approval of the ADC trastuzumab deruxtecan for HER2-positive cancers underscores the value of orthogonal cell-surface protein targeting that might bypass the limitations of small molecule-induced genomic complexity. For all of the above reasons, the development of clinical-grade comprehensive proteomic assays is an unmet need.

Most importantly, maximizing the likelihood of cure is essential. Insights on MRD will help us to understand how to treat drug-tolerant persister cells. Additionally, although ICIs have limited activity in many driver-positive lung cancers⁵⁰, the hope is that novel cellular and non-cellular immunomodulatory therapies will turn these historically cold tumours hot.

L.G.:

For patients with cancer, the risk of overtreatment or useless treatment is large. For patients potentially cured of breast cancer in particular, de-escalation of therapy is an emerging priority. The neoadjuvant period is already used to make treatment decisions when residual disease is detected^{15–17} and can provide clues for individual tailoring of treatment beyond de-escalation. For example, many patients could already be cured with locoregional interventions only, and special effort should be made to identify such patients and spare them from useless additional treatment. Formal validation of pathological complete response

as a surrogate of potential cure together with the already available validation of residual disease as a marker of resistance could shape new strategies based on response-adapted treatment and on markers of residual disease during and after neoadjuvant therapy as initial trials such as PHERGain are illustrating⁵¹.

J.T.:

Early-onset cancer is an emerging global epidemic⁵². We must decipher why the number of young adults diagnosed with cancer is increasing, and identify specific early-life exposures (including smoking, diet, lifestyle and alcohol) that contribute to the rising incidence of CRC and other cancer types in young adults. The exposome can cause (epi)genetic alterations in cancer cells and modulate the gut microbiota and host immunity. We need to gather more data from large populations and prospective cohort analyses to better understand environmental cancer risk as well as devise earlier detection and improved prevention strategies.

The study of gut microbiota is a fascinating field in oncology because microbial communities have a key role in drug metabolism, efficacy and toxicity. These communities can influence an individual's response to a given treatment and act as codrivers of cancer development.

H.I.S.:

Moving forward, prostate cancer research must focus on understanding the key pathways that induce epithelial plasticity and other resistance mechanisms using single-cell RNA sequencing and proteomic techniques. Identifying biomarkers that can detect these changes before clinical manifestation is crucial as is developing effective treatments for these highly heterogeneous neuroendocrine phenotypes that now account for 20% of advanced-stage prostate cancers. The identification and profiling of the specific cellular populations present in patients with this disease will be key to deconvoluting their heterogeneity and sequentially eliminating them.

J.M.L.:

The future management of patients with HCC looks promising. First, neoadjuvant and adjuvant therapies will flourish for those with early-stage disease and are expected to improve outcomes, similar to the results observed in patients with melanoma or NSCLC⁵³. Second, phase III trials in the advanced-stage setting are testing triplet regimens in which a third drug (such as an anti-TIGIT or anti-LAG3 antibody) is added to ICI-ICI, and ICI-anti-VEGF antibody or TKI combinations. Triplet regimens could enable a wider and deeper effect in patients who currently have disease control in response to doublet regimens. Similarly, personalized vaccines could further boost immune responses by activating T cells with novel neoantigens. Third, once these efforts reach a ceiling of clinical benefit or unacceptable toxicities, the development of biomarkers to maximize benefit from precision oncology should be prioritized. Nonetheless, one-third of patients with advanced-stage HCC still present with an aggressive phenotype that remains primarily resistant to ICIs owing to a strong immunosuppressive TME. For these patients, drugs abrogating pathways involved in immunosuppression (such as WNT- β -catenin, Notch or TGF β) or, alternatively, CAR T

cells or CAR macrophages targeting common HCC-related antigens might emerge as novel approaches.

S.B.:

The top priorities in gynaecological oncology are prevention, earlier diagnosis and increasing cure rates. Continued successful implementation of human papillomavirus vaccination programmes and screening worldwide is necessary for prevention of cervical cancer, whereas the quest for effective screening strategies for ovarian cancer continues. Therapeutic approaches tackling resistance to PARP inhibitors in patients with ovarian cancer, and to ICIs in those with endometrial or cervical cancer, are required. Finally, treatment advances for patients with rare gynaecological cancers remain a priority.

Challenges for the next 20 years include how to integrate the wealth of molecular information guiding treatment decisions and how to best sequence biomarker-directed therapies such as ADCs. Research on treatment-related toxicities — mechanisms, prediction, novel management strategies and risk stratification across acute, long-term and late-onset effects — is needed. Our vision is that the number of long-term survivors of gynaecological cancer will increase in the next two decades owing to progress in surgery, radiotherapy and systemic therapies.

M.W.B.:

In the next 20 years, we must focus on advances towards more precise surgery and systemic therapies based on a deeper understanding of tumour plasticity. For patients with pancreatic cancer in particular, the most disappointing outcome after surgery is local recurrence, which can occur in as many as 30–40% of patients. This observation indicates that the type of surgery performed, even in specialized centres, is not sufficient for local clearance. Pancreatic cancer tends to spread via lymphatic and neural networks, requiring complete technical clearance of all local tumour material. This need has led to the development of the Triangle Operation⁵⁴, which involves integral venous and arterial vascular resections in a personalized approach. Depending on the location (pancreatic tail, body, head or uncinate process) and the individual cancer extension outside of the pancreas, different vascular, lymphatic and neural structures must be approached individually. Initial clinical trials applying a precise surgical approach to prevent local recurrence have shown that, with such a multimodal concept, better local cancer clearance leads to improved OS in patients with pancreatic cancer.

M.W.:

The major challenge in neuro-oncology, the management of diffuse gliomas, has seen two waves of great hope and even greater disappointments: first, the wave of anti-angiogenic agents that left us with a much-debated approval for bevacizumab for patients with recurrent glioblastoma in some regions of the world, and later, a series of negative phase III trials testing cancer vaccines or ICIs. Based on these experiences, future advances will probably be driven by smaller but much better-designed clinical trials, including phase 0 clinical trials designed to verify target engagement, and focusing on biomarker-selected patient populations. Moreover, local therapies, including ACT with CAR T cells, might gain

momentum because these approaches reduce systemic toxicity and glioblastoma remains a disease essentially restricted to the central nervous system. We have traditionally assumed that the lack of metastases from primary brain tumours could be related to superior immune control outside of, as opposed to within, the brain. The emerging field of cancer neuroscience, which has identified intricate interactions between brain parenchymal cells and brain cancer cells, provides another explanation of why the growth of these tumours is primarily local. Interventions targeting this potentially essential crosstalk between brain and cancer cells open up a new avenue for brain cancer-specific pharmacological interventions⁵⁵.

K.P.-J.:

A top priority in paediatric oncology is to expand the evidence base for the use of novel therapies, including an improved understanding of epigenetic control of developmental pathways. 'Big data' approaches to recording adverse events outside of clinical trials are also needed, recognizing that it is essential to monitor throughout a child's normal physiological development and growth. We also need to foster research in low-income and middle-income countries, where >90% of the world's children live, because the early adoption of newer therapies might be better tolerated and improve efficacy⁵⁶. Innovations in earlier diagnosis, including circulating biomarker approaches, and their application to achieve more accurate assessment, are global priorities, particularly in low-income and middle-income countries. Research in cancer prevention, starting with individuals with a known genetic predisposition, is another priority. We must empower cancer survivors to understand their modifiable risk factors to live with good QOL. Finally, we must involve patients and families affected by childhood cancer in defining the research priorities that matter most to their communities, leading to better-designed and higher-quality trials and studies that are more likely to succeed and thus change practice.

R.R.W.:

Future research in radiotherapy should focus on clinical and molecular cancer staging and on use of ctDNA to identify patients with limited disease who are most likely to benefit from this treatment modality⁵⁷. Preclinical and clinical research into the mechanisms of action of radiotherapy alone or combined with targeted agents, hormonal agents, immunotherapies and cytotoxic agents are essential to improving outcomes. The combination of theranostic agents with radiotherapy and systemic agents to enhance cancer cell killing and guide ablative interventions might broaden the treatment window to include patients who have more extensive disease, thereby increasing the proportion of patients with metastatic cancer who can benefit from these treatments.

K.P.:

Future clinical applications of liquid biopsies include early detection of both primary cancer and disease relapse, improved cancer staging, real-time monitoring of therapeutic efficacy, and detection of therapeutic targets and resistance mechanisms. To reach this ambitious goal, initiating more interventional trials to demonstrate the clinical utility of liquid biopsies is an important prerequisite for the introduction of this new technology into clinical practice. Moreover, assay harmonization and standardization efforts from international consortia (such as the European Liquid Biopsy Society) are essential.

Early detection and molecular characterization of MRD (or molecular residual disease) could be the most innovative of the aforementioned clinical applications, potentially opening a new avenue for adjuvant and post-adjuvant therapies⁵⁸ aimed to prevent or slow down the progression to overt metastatic relapse. In this context, multiomics assessment of liquid biopsy samples could provide unique information on targetable alterations in both cancer and host cells with relevance for MRD control.

E.B.:

The next 20 years will determine what works in the ‘palliative care syringe’. We need a better understanding of the ideal type and dose of palliative care interventions given the limited number of clinical palliative care programmes routinely available. Emerging academic palliative programmes will develop better evidence-based interventions to mitigate pain, fatigue, delirium, cachexia, dyspnoea, and emotional distress and improve patient–clinician communication. These advances will be possible through the complete and seamless integration of palliative care within cancer centres and universities.

C.M.B.:

In the next 20 years, we need to address the current barriers to delivering high-quality and equitable cancer care, with an increased focus on value-based pricing and health technology assessment. The most pressing research priority in the next two decades is to perform mixed-methods research to better understand how patients perceive these complex cost–benefit trade-offs and how we can better deliver outcomes that matter⁵⁹. Ultimately, the oncology clinical and research communities need to have frank discussions about where we can put our efforts to have the greatest impact on the lives of people who have cancer.

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